

Operating Systems & Systems Programming

SITAMS CSE (AI & ML) • R23 • III Year • VI Semester • Course Code: 23CSE353T

Source basis: Official SITAMS R23 CSE (AI & ML) syllabus uploaded for this Study Hub. This concise revision guide preserves the syllabus topic statements and adds a study checklist; it is not a replacement for prescribed textbooks or faculty notes.

Unit 1: FUNDAMENTALS OF OPERATING SYSTEMS AND PROCESS MANAGEMENT

Syllabus topics

- Introduction to Operating Systems: Definition and Basics, Generations and Types of Operating Systems, OS Structure: Layered, Monolithic, Microkernel, OS Services, System Calls, System Boot, System Programs, Virtual Machines, Process Management: Process Concepts, Process States, Process Control Block, Context Switching, Threads and Multithreading, Process Scheduling: Scheduling Criteria and Scheduling Algorithms, Multiprocessor Scheduling: Types and Performance Evaluation, Process Synchronization and Deadlocks: Race Conditions, Critical Section, Mutual Exclusion, Peterson's Solution, Semaphores, Monitors Classic IP, C Problems: Reader-Writers, Dining Philosophers, Deadlocks: Definition, Characteristics, Prevention, Avoidance, Detection and Recovery

Revision checklist

- Know the definitions and principles named in the syllabus.
- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.
- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

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Unit 2: MEMORY, FILE, AND STORAGE MANAGEMENT

Syllabus topics

- Memory Management: Logical vs
- Physical Address Mapping, Contiguous Memory Allocation, Internal and External Fragmentation, Compaction, Paging and Page Tables, Segmentation, Virtual Memory: Demand Paging, Page Faults, Page Replacement Algorithms, Thrashing and Working Set Model, File System Management: File Concepts, Access Methods, File Types and Operations, Directory Structure, File System Structure, Allocation Methods, Free-Space Management, Directory Implementation
- Storage Management: Mass Storage: Disk Structure, RAID Levels, Disk Scheduling Algorithms, Swap Space Management, Stable Storage, Tertiary Storage Structure

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Unit 3: I/O SYSTEMS, SECURITY, AND UNIX/LINUX OVERVIEW

Syllabus topics

- I/O System Management: I/O Hardware: Devices, Device Controllers, Direct Memory Access, I/O Software: Interrupt Handlers, Device Drivers, Device-Independent I/O Software, System Protection and Security: Security Environment, Security Design Principles, User Authentication, Protection Mechanisms, Protection Domain, Access Control List, Unix/Linux Overview & Case Studies: Development of Unix/Linux, Role of Kernel, System Calls, Elementary Linux

Commands, Shell Programming, Directory Structure, System Administration

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Unit 4: SYSTEM SOFTWARE AND LANGUAGE PROCESSING

Syllabus topics

- Overview of System Software: Software and Software Hierarchy, Systems Programming and Machine Structure, Interfaces, Address Space, and Computer Languages, System Software Development and Recent Trends, Language Processors: Programming Languages and Language Processing, Symbol Tables and Data Structures for Language Processing, Search and Allocation Data Structures, Assemblers and Macro Processors: Elements of Assembly

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Unit 5: SYSTEM PROGRAMMING

Syllabus topics

- Scanning and Parsing: Programming Language Grammars and Classification, Ambiguity in Grammatical Specification, Scanning, Parsing, Compilers and Interpreters: Compilation Process, Semantic Gap, Binding, and Scope Rules, Memory Allocation, Compilation of Expressions & Control Structures, Code Optimization, Overview of Interpreters and Debuggers, Operating System Command & Shell Basics: C Development Tools, Machine- Level Representation of Data and Programs, System-Level Programming and Concurrency: File I/O, Process Creation & Control (fork, exec), Pipes, Signals, and Basic Threading

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Study Hub status

This guide is syllabus-derived. External resources and quiz questions are tracked separately and are only labelled verified after source checking.